

LUT core-module ablation: three generations of lookup math

Comparison table

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Protocol

Architecture: LUT transformer, $E = 384$, 6 layers, 6 attention heads. FFN = **CompressionMultiHeadLUT** with $H = 4$ heads $\times$ $tph = 128$ tables ($P = H \cdot tph = 512$ tables per token), $d_{in} = d_{out} = 48$, $n = 8$ anchor pairs per table ($K = 2^n = 256$ cells); **compress** $384 \rightarrow 192$, **decompress** $192 \rightarrow 384$. **Training:** 16,000 steps, batch size 48 (24576 tokens), untied embeddings, LUT tables without weight decay. **Eval:** **evaluate_bpb_fixed**, 48 rows $\times$ 100 steps, first 12 rows skipped.

One seed per row. Every validation bpb below is a single run, and there are **zero seed replicates** at this geometry. The only LUT seed-noise figure on record is a 4K three-seed sd of 0.009642, a *lower bound*; “units” below are multiples of it (the vanilla two-seed range at 16K is 0.00335). **Every Gen-3 difference in Table B is inside one unit:** 3.2 vs 3.1 -0.000016 (< 0.01), 3.3 vs 3.2 -0.0056 (0.6).

Counting convention

- Core LUT only, per token, one layer. The **compress/decompress** projections are identical in every row and excluded (147,456 MACs forward).
- **FLOP** = one scalar float operation ($+$, $-$, $\times$, $\div$, a comparison, $|\cdot|$, $\exp$, $\log \sigma$, σ). **MAC** = a multiply accumulated into a sum; 1 MAC = 2 FLOPs, included in the FLOP column. Memory reads and integer work are not counted.
- **Forward** = inference. **Backward** = all additional training arithmetic, including per-table gradients of per-layer scalars and training-only regularisers (amortised over the 24,576 tokens of a step).

Table A — forward and backward math

Variant	Forward pass (math)	Backward pass (math)
GEN 1 — Spiking Manifesto basic model^a		
1.1 Hard forward, two-alternatives backward	$y_h = \sum_{t \in h} W_t[c_t]$: head h sums, over its tables t , the row of $W_t \in \mathbb{R}^{K \times d}$ at cell c_t , the bits $[u_j > 0]$ of the n margins $u_j = z_h[a_j] - z_h[b_j]$.	Weights: $\partial L / \partial W_t[c_t] += g$, $g = \partial L / \partial y_h$. Input (surrogate: the derivative of 1.2's blend), least-confident pair $j^* = \arg \min_j u_j $ only: $\partial L / \partial u_{j^*} = -U'(u_{j^*}) (G_t(c_t) - G_t(c'_t))$, with $U(u) = 0.5 / (1 + u)$, $-U'(u) = 0.5 \operatorname{sgn} u / (1 + u)^2$, $G_t(\ell) = \langle g, W_t[\ell] \rangle$, and $c' = c$ with bit j^* flipped.
1.2 Soft forward, two-alternatives backward	$y_h = \sum_t [(1 - U_t) W_t[c_t] + U_t W_t[c'_t]]$, $U_t = U(u_{t,j^*})$; continuous, applied at eval too.	Weights: $(1 - U_t) g$ into $W_t[c_t]$, $U_t g$ into $W_t[c'_t]$. Input: as 1.1, here the exact gradient of the blend with both cells fixed.
GEN 2 — FastMHL math^b		
2.1 Hard forward, all-alternatives backward (all K cells)	$y_h = \sum_t W_t[c_t]$, $c_t = \arg \max_\ell s_\ell$, cell score $s_\ell = \sum_j \pm \rho_j$ (+ where bit j of ℓ equals $[u_j > 0]$), $\rho_j = u_j / (T_s + u_j)$.	Weights: $\partial L / \partial W_t[c_t] += g$. Input, via the surrogate $\sum_\ell \pi_\ell W_t[\ell]$, $\pi_\ell = \operatorname{softmax}_\ell(s_\ell / T_{sel})$ (not used in the forward; reads all K rows; T_s, T_{sel} temperatures): $\partial L / \partial u_j = \sum_\ell \pi_\ell (G_t(\ell) - \bar{G}_t) (\partial s_\ell / \partial \rho_j) T_s \operatorname{sgn} u_j / (T_{sel}(T_s + u_j)^2)$, with $\bar{G}_t = \sum_\ell \pi_\ell G_t(\ell)$ and $\partial s_\ell / \partial \rho_j = \pm 1$.
2.2 Hybrid_smooth two-alternatives forward, all-alternatives backward	$y_h = \sum_t [(1 - w_t) W_t[c_t] + w_t W_t[c'_t]]$, $w_t = \sigma(-2\rho_{t,j^*} / T_{sel})$, σ the logistic sigmoid; applied at eval too.	Weights: $(1 - w_t) g$ and $w_t g$. Input: the surrogate of 2.1.
2.3 Hard forward, 2-alternatives backward	$y_h = \sum_t W_t[c_t]$	TBD — not implemented; open decision. FastMultiHeadLut raises <code>ValueError</code> for <code>backward_topk>0</code> unless <code>forward_mode='hybrid_smooth'</code> .
2.4 Hybrid_smooth two-alternatives forward, 2-alternatives backward	As 2.2.	Weights: as 2.2. Input (<code>backward_topk=1</code>), the exact gradient of the 2-cell blend, j^* only: $\partial L / \partial u_{j^*} = -w_t(1 - w_t) (G_t(c'_t) - G_t(c_t)) 2T_s \operatorname{sgn} u_{j^*} / (T_{sel}(T_s + u_{j^*})^2)$.
GEN 3 — LightMHL math (LookupFFN-inspired): learned margin score^c		
3.1 Learned margin score, per-layer (g, β, γ)	$y_h = \sum_t s_t W_t[c_t]$, $s_t = e^g S_t P_t^\gamma$, $S_t = \sum_j m_j$, $P_t = \prod_j \sigma(\beta m_j)$, $m_j = u_j $; computed as $S_t \exp(g + \gamma \sum_j \log \sigma(\beta m_j))$. One ($g, \log \beta, \log \gamma$) per layer, initialised at $(0, 2, 1)$; the cell index uses detached u . Discontinuous at cell boundaries, since $\sigma(0) > 0^d$.	Weights: $s_t g$. Input: $\partial L / \partial u_j = G_t(c_t) (e^g P_t^\gamma + \gamma \beta s_t \sigma(-\beta m_j)) \operatorname{sgn} u_j$. Per layer: $\partial L / \partial g = \sum_t G_t(c_t) s_t$, $\partial L / \partial \log \beta = \sum_t G_t(c_t) s_t \gamma \sum_j \beta m_j \sigma(-\beta m_j)$, $\partial L / \partial \log \gamma = \sum_t G_t(c_t) s_t \gamma \log P_t$.
3.2 The Gen-3 form: learned margin score, per-layer (β, γ)	As 3.1 with g fixed at 0: $s_t = S_t P_t^\gamma$.	As 3.1 without $\partial L / \partial g$.
3.3 = 3.2 + Hamming-1 cell TV, $\lambda = 10$	Identical to 3.2 (the penalty is training-only).	As 3.2, plus once per step the gradient of $\lambda \cdot \frac{1}{6} \sum_{\text{layers}} \frac{1}{P n 2^{n-1}} \sum_t \sum_{(c, c')} \ W_t[c] - W_t[c']\ ^2$ over cell pairs differing in one bit: $\partial L / \partial W_t[c] += \frac{2\lambda}{6 P n 2^{n-1}} \sum_{c' \sim c} (W_t[c] - W_t[c'])$ for every cell.

Notes to Table A.

^a The code also offers `UncertaintyMode.INVERSE_QUADRATIC`, $0.5 / (1 + u^2)$.

^b **Config trap:** `backward_topk` counts cells *beyond* the addressed one. “2 alternatives” is `backward_topk=1`, not 2; “all alternatives” is `backward_topk=0`.

^c A hard-forward Gen-3 variant has not been designed.

^d Measured boundary jump, median share of $\|y_h\|$ on trained weights: 3.1 2.19%; 3.2 2.22%; 3.3 1.78%.

Table B — cost and results

Per token, one layer, core LUT only; $P = 512$, $n = 8$, $K = 256$, $d = 48$. Validation bpb: **one seed per row**.

Variant	Forward FLOPs	Backward FLOPs	Forward MACs	Backward MACs	Validation bpb	Implementation (class used)
1.1	$P(2n + d) = \mathbf{32,768}$	$P(5d + 2n + 9) = \mathbf{135,680}$	0	$P \cdot 2d = \mathbf{49,152}$	TBD — not yet implemented	MultiHeadLut(smooth_mode=False, n_alternatives=1, uncertainty_mode=INVERSE_L1); not wired into CompressionMultiHeadLUT.
1.2	$P(4d + 4n + 4) = \mathbf{116,736}$	$P(8d + 9) = \mathbf{201,216}$	$P \cdot 2d = \mathbf{49,152}$	$P \cdot 4d = \mathbf{98,304}$	TBD — not yet implemented	MultiHeadLut(smooth_mode=True, n_alternatives=1); not wired in.
2.1	$P(2n + d) = \mathbf{32,768}$	$P(2Kd + 4nK + 9K + 11n + d) = \mathbf{18,026,496^e}$	0	$P(Kd + 2nK) = \mathbf{8,388,608}$	1.180622 (exp_n_0121) ^f	FastMultiHeadLut(forward_mode='hard', backward_topk=0).
2.2	$P(2kd + 4n + 5) = \mathbf{117,248}$, $k = 2$ cells read	$P(2Kd + 4nK + 9K + 11n + 2kd) = \mathbf{18,100,224^e}$	$Pkd = \mathbf{49,152}$	$P(Kd + 2nK + kd) = \mathbf{8,437,760}$	TBD — not yet run	FastMultiHeadLut(forward_mode='hybrid_smooth', backward_topk=0).
2.3	$P(2n + d) = 32,768$	n/a	0	n/a	TBD — not implemented; open decision	Would be FastMultiHeadLut(forward_mode='hard', backward_topk=1), which the code rejects.
2.4	$P(2kd + 4n + 5) = \mathbf{117,248}$	$P(4kd + 9n + 15) = \mathbf{241,152}$	$Pkd = \mathbf{49,152}$	$P2kd = \mathbf{98,304}$	TBD — not yet run	FastMultiHeadLut(forward_mode='hybrid_smooth', backward_topk=1) (note b).
3.1	$P(2kd + 7n + 2) = \mathbf{78,848}$, $k = 1^g$	$P(4kd + 10n + 10) = \mathbf{144,384^g}$	$Pkd = \mathbf{24,576}$	$P2kd = \mathbf{49,152}$	1.169344 (exp_g_0247)	LightMultiHeadLUT(confidence_form='learned_margin', learned_margin_init=(0, 2, 1)).
3.2	$P(2kd + 7n + 2) = \mathbf{78,848}$	$P(4kd + 10n + 8) = \mathbf{143,360^g}$	24,576	49,152	1.169328 (exp_g_0248)	As 3.1 with learned_margin_freeze_g=True.
3.3	78,848 (as 3.2)	149,504 (3.2 + TV) ^h	24,576	51,200^h	1.163698 (exp_g_0249)	As 3.2, plus (10 * model.lut_tv_penalty()).backward() per step (lut_cell_smoothness: 10.0).

Notes to Table B.

^e Dominated by $2Kd$: every table's K rows are read; several [tokens, 512, 256] fp32 buffers of about 12.9 GiB at 24,576 tokens.

^f Re-score of the saved run under the corrected eval (0 missing keys); its in-run batch-coupled number was 1.191455.

^g Per table, forward: n comparisons, n differences, n absolute values, n products βm_j , $n \log \sigma$, the two sums, $\times \gamma$, $+g$, exp and the final product $(7n + 2)$, plus the scaled row $(2kd)$. Backward: the row terms $(4kd)$, the input term $(8n + 3)$ and the per-table accumulation of the parameter gradients $(\partial g_2, \partial \log \gamma_3, \partial \log \beta_2n + 2)$; 3.2 drops ∂g . The parameter gradients are scalar multiplies, so they add no MACs.

^h Training only, once per step: $Pn2^{n-1} = 524,288$ cell pairs per layer, $3d$ FLOPs and d MACs per pair in each direction (150,994,944 FLOPs, 50,331,648 MACs per layer per step), i.e. +6,144 FLOPs and +2,048 MACs per token over 24,576 tokens. Inference cost is 3.2's.

The Gen-3 form (one seed each)

- **Learned margin score with per-layer trainable β and γ (3.2).** Drop g : freezing it leaves bpb (-0.000016 vs 3.1) and the γ profile ($\max |\Delta\gamma|$ 0.016) unchanged. Trained γ 1.04 – 1.60 , highest mid-depth; β 1.76 – 2.17 and nearly irrelevant to the score’s shape. Both were still drifting at $16K$.
- **Cell TV (3.3): -0.0056 vs 3.2 , a mild regularisation benefit**, not established (0.6 units). **The mechanism is mostly table shrinkage, not smoothing:** table L2 norms fell to 0.22 – $0.35\times$ of 3.2 ’s, while the scale-free ratio $\|W[c] - W[c']\|^2 / \|W[c]\|^2$ (2.0 for independent random cells) moved only from 1.74 – 1.95 to 1.34 – 1.57 across layers.

Where the code differs from the original brief

- 1.1 ’s input gradient is a surrogate (the derivative of 1.2 ’s blend); its weights update only the addressed row.
- The status report’s $u(\delta) = 0.5/(T + |\delta|)$ has a temperature; the code’s $0.5/(1 + |u|)$ has none (`l_projection.py:96`, `lutorch.cu:797`).
- Gen 2 ’s `hybrid_smooth` weight $\sigma(-2\rho_{j^*}/T_{sel})$ is a temperature-controlled soft-sign, not Gen 1 ’s $U(u)$.

Remaining work

- **Run 2.2 and 2.4:** fork `exp_n_0121` verbatim, changing only `lut_forward_mode` / `lut_backward_topk`. No matching run exists (earlier `hybrid_smooth` runs are $n = 6$, tied, old eval; `exp_n_0188` is $n = 7$ with a gate on).
- **Implement 1.1, 1.2:** wire Gen 1 into `CompressionMultiHeadLUT`, which accepts only `lut_impl` $\in \{\text{fast, light, bh4}\}$ (`compression_mhl.py:190`), as `MultiHeadLut` with `anchor_candidates` $[128, 4, 48]$. **Open decision, 2.3:** patch `FastMultiHeadLut` (`fast_multi_head_lut.py:1417`) or drop the row.
- **Seeds:** zero replicates at this geometry; one extra seed of `exp_g_0248` (≈ 0.95 h) would give a real LUT seed spread. Reference: vanilla dense FFN 1.1651 / 1.1618 at $16K$ (two seeds), 1.1154 at $48K$.